MUHAMMAD ANAS RAZA

Industrial Engineering | Manufacturing Methods | Continuous Improvement

Grenoble, France | +33 7 54 86 45 54 | muhammad-anas.raza@grenoble-inp.org
linkedin.com/in/muhammadanas-raza-58060617a | Engineering portfolio ↗

Six-month internship from February 2027 | Mobile across France



SUMMARY

Mechanical engineer pursuing M2 Sustainable Product Development at Grenoble INP–UGA, following M1 Sustainable Industrial Engineering. Experience in shop-floor improvement at Hprobe, machining quality, Python KPI dashboards, discrete-event simulation and hands-on prototyping.

SKILLS

Methods & quality: VSM, 5S, DMAIC, 8D, FMEA/AMDEC, 5 Whys, Ishikawa, root-cause analysis
Data & simulation: Python (KPI dashboards), Excel, Power BI, R, FlexSim Healthcare, OEE, first-pass yield
Design & industrialisation: SolidWorks, ANSYS, mechanical prototyping, technical drawings, tolerance analysis
Sustainability & operations: SimaPro, LCA (ISO 14040/44, ReCiPe 2016, Impact World+), ramp-up, risk tracking

PROFESSIONAL EXPERIENCE

Continuous Improvement & Industrial Organisation Intern *May–Aug 2026*
Hprobe Eybens, France

- Mapped assembly flows using VSM and identified three bottleneck stations; contributed to a reported **18% reduction in WIP**.
- Supported 5S improvements across two production cells, associated with a reported **12% reduction in setup time**.
- Built a Python dashboard tracking OEE, first-pass yield and cycle time for daily production reviews.
- Used DMAIC, 5 Whys, Ishikawa and FMEA to investigate assembly defects; helped production and R&D teams track line ramp-up actions.

Manufacturing Engineering Intern – Transmission Shafts *Sep–Dec 2024*
Nanjing Donghua Driving Shaft Ltd. China

- Made defects and rework traceable by batch and machine using a Python tracking tool.
- Analysed **2 recurring defects** and tolerance issues across **6 shaft designs** using 5 Whys and FMEA.

Research Intern – Composite Materials *May–Aug 2024*
Nanjing Tech University China

- Evaluated composite stiffness, deformation and failure in SolidWorks and ANSYS, comparing simulations with theory and test data.

Technical Sales Manager (Remote) *Jan–Jul 2024*
Shanghai Youwei Powder Equipment Co. China

- Supported **20+ clients across 5+ markets** with equipment quotations, technical documents and supplier coordination.

PROJECTS

Filament-to-Pellet Slicer | Grenoble INP FabLab *2025–2026*

- Designed the feeding and removable collection mechanisms; helped build and test a working prototype reaching **17.6 pellets/s** and **2.06 mm** average pellet length (team test).

Emergency Department Optimisation | FlexSim Healthcare *2026*

- Simulated three emergency-department demand scenarios; proposed capacity changes modelling **328 to 683 consultation/treatment events/day** at **43% lower estimated cost per patient**.

Comparative Battery Life Cycle Assessment | Grenoble INP *2025*

- Modelled alkaline versus rechargeable AAA battery impacts in SimaPro using Impact World+ and ReCiPe 2016, accounting for production, use, transport and end of life.

EDUCATION

M2 – Industrial Engineering, Sustainable Product Development (SPD) *2026–2027 (expected)*
Grenoble INP–UGA, France

M1 – Sustainable Industrial Engineering *2025–2026*
Grenoble INP–UGA, France *IDEX scholarship recipient | M1 GPA: 14/20 | Top 3 in class*

B.Eng. – Mechanical Engineering *2021–2025*
Nanjing Tech University, China *CGPA: 3.32/4.00*
Mechanical design, manufacturing, materials and structural analysis; graduation thesis rated **Excellent**.

PUBLICATION

Co-author, "Revolutionizing Two-Dimensional Pattern Arrangement: Innovative Image Processing Approaches," *IEEE Access*, vol. 12 (2024). DOI: 10.1109/ACCESS.2024.3447092.

LANGUAGES

English (C2) | Mandarin (B2) | French (A2/B1, improving) | Urdu/Hindi (native)